Observe, Imagine, Abstain: Detecting Hallucinated Geometry in Feed-Forward 3D Reconstruction

Anonymous CVPR submission

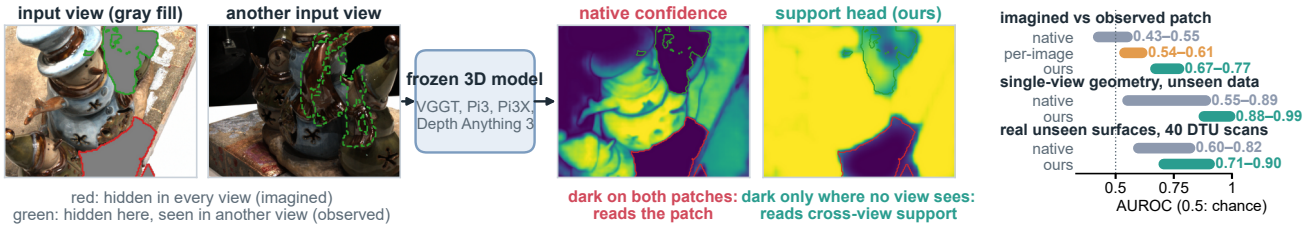


Figure 1. **Native confidence reads the patch, not the missing support.** Two identical gray patches: red is hidden in every input view (imagined), green is hidden here but visible in another view (observed). Native confidence is dark on both; our label-free support head, which reads the backbone's cross-view tokens, is dark only where no view sees the surface. Right: AUROC ranges over backbones and datasets for the matched patches, for single-view geometry on four unseen datasets, and for surfaces no input view saw on 40 real DTU scans.

Abstract

Feed-forward 3D models reconstruct a scene from unposed photographs in one pass. Where no photograph shows a surface they invent one, and report it with the confidence of an observed surface. We show that this confidence does not know what the model saw: against an identical patch that another view does see, it is at chance on five backbones (AUROC 0.43 to 0.55), and it lets 24% to 81% of invented geometry through a filter that keeps 90% of observed geometry. The model's cross-view tokens do know. A lightweight support head, *trained on the backbone's own multi-view agreement with no ground truth, flags single-view geometry at AUROC 0.88 to 0.99 on four datasets it never saw (native confidence: 0.55 to 0.89). On 40 real DTU scans, against a dense reference that records what every camera saw, it finds surfaces no input view observed better than native confidence in 14 of 16 settings, on four backbones. A larger transformer head reads no more: the signal is already in the features.*

1. Introduction

Feed-forward 3D reconstruction has moved from pairwise pointmaps [14, 31] to models that take dozens of unposed images and return dense geometry in one pass [13, 17, 29, 32]. They output geometry at every pixel, including pixels whose surface no input view shows: the objective rewards a plausible guess, so they learn to guess. A conservator who sees a crack in a reconstructed dome needs to know whether a photograph shows it or the model drew it from what domes look like. Today the confidence map is bright in both cases.

What we find.

1. **Confidence does not know what the model saw.** A quarter to four fifths of the invented geometry passes a confidence filter, and against an identical patch that another view sees, confidence is at chance.
2. **The cross-view tokens do.** A linear probe on them reaches AUROC 0.99; a head trained without labels reads the same signal on four backbones and four unseen datasets, and a larger transformer head reads no more.
3. **It holds on real scans.** On 40 DTU scans, against a dense reference, the head beats native confidence on surfaces no input view saw in 14 of 16 settings, by up to 0.14 AUROC.
4. **A consistent invention passes every cross-view test,** ours included, and five models invent the same surface. It follows the pixels it was drawn from.

Contributions. (i) A definition of geometric hallucination and HalluBench-3D, with a *matched split* that removes the patch confound and makes a per-image control a required test for any detector of unobserved geometry (Sections 3

and 4). (ii) A label-free *support head* on the frozen backbone that abstains between observed and imagined (Figure 5, Section 5), tested on four backbones and four datasets against native confidence, reprojection, a per-image control and a ground-truth-trained head (Section 6). (iii) A real-occlusion test against dense references on 40 DTU scans and 18 CO3Dv2 phone captures, a capacity test from a linear probe to a cross-view transformer, and the limit of every cross-view test (Section 6). Every threshold and pass condition was fixed before the run it judges.

2. Related work

Feed-forward 3D. DUSt3R and MASt3R [14, 31] predict pointmaps for image pairs; VGGT [29], Pi3 and Pi3X [32], MapAnything [13], VGGT-Omega [30] and Depth Anything 3 [17] take many unposed views in one pass. Each exposes a per-pixel confidence trained as a loss weight on observed data; none is trained to tell observed from unobserved surfaces.

Hallucination in feed-forward 3D. Attention heads of DUSt3R, VGGT and DA3 keep matching correspondences after small patches are occluded [3], measured with attention and Sampson error rather than confidence. Unrelated scenes, repeated images and noise still produce dense geometry [23]. Monocular depth fails on illusions [21]; wide viewpoint spans produce unsupported geometry [15]; unobserved parts of a room escape inference-time uncertainty [5]; inpainting fills masks on purpose [22]. None has per-point labels. In vision-language models POPE [16] labels statements the image does not support; ours is the geometric analogue.

Uncertainty for feed-forward 3D. ReVar3R [18] takes per-point variance over perturbed runs and, by its authors' account, misses stable systematic bias, which a confidently filled region is. Trust3R [33] fits an evidential head, GeoCond [1] adds pose-level uncertainty, and Deng et al. [7] find the implied uncertainty of seven backbones at least 2.4× too small. All score uncertainty against error, which conflates confident errors on observed surfaces with confident predictions on unseen ones; CUR isolates the second. Closest to our detector, a linear probe on frozen VGGT features predicts camera-pose failure [28] and a small head predicts pairwise co-visibility [34]; our head is the dense, per-pixel analogue for unobserved *geometry*, trained with no label at all.

Generative 3D. Hidden surfaces are decodable from frozen VGGT features [2]; feed-forward splatting [9, 26] and world models [4] store imagined and observed surfaces alike; sample disagreement [11, 12, 20] and quantized generative heads [6, 19] are the natural successors to a discriminative head.

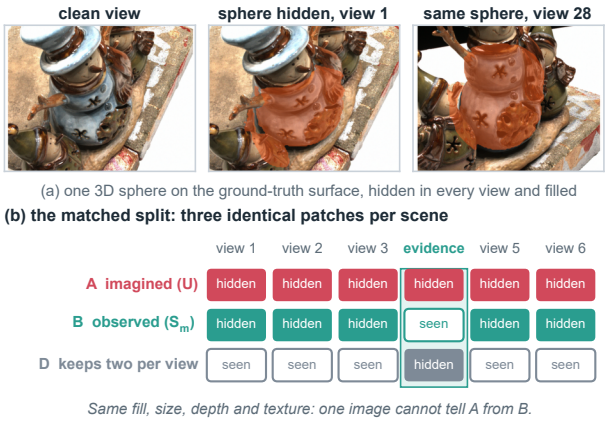


Figure 2. **The benchmark and its matched split.** (a) A sphere on the ground-truth surface is hidden in every view, so no camera sees it; its pixels are unsupported. (b) The matched split adds identical patches that one evidence view still sees, so only cross-view evidence separates imagined from observed.

3. HalluBench-3D

Definition. A model returns depth D_i and confidence C_i for each input view i. A pixel is *supported* if its ground-truth surface point is visible in at least one input view and *unsupported* otherwise; a model hallucinates when it outputs confident geometry at an unsupported pixel. Native confidence is a loss weight learned on observed data, so a smooth invented surface is confident by construction.

Metrics. AUROC is $P(C_S > C_U)$, with 0.5 at chance. The Confident-Unsupported Rate CUR sets τ so that 90% of supported pixels pass and counts the unsupported pixels that pass too: the hallucination rate after filtering. CUR-ERR adds a depth error above 10%. Means carry 95% cluster-bootstrap intervals over scenes; comparisons use paired sign-flip tests on per-scene means, Holm-corrected ($p < 0.0001$ is the test's floor).

Protocol. Ten views per scene. A sphere of radius 0.10 or 0.18 of the object diagonal is centred on the ground-truth surface and every pixel whose ground-truth point lies inside it is hidden in every view (Figure 2a) and filled with gray, a blur, noise or an inpainting [27]. Hidden pixels are U, pixels outside a 6-pixel margin are S. Five seeds per scene.

The matched split. Every U pixel sits under a patch and no S pixel does, so a confidence that merely reacts to a patch would look as if it knew about missing support. The matched split (Figure 2b) places three spheres matched in area, depth, texture and fill: A is hidden in every view (U); B is hidden except in an evidence set of one, two or half the views, so its pixels are filled but observed (S_m); D is hidden only in the evidence views, so every view carries two patches. AUROC$_m$ and CUR$_m$ score U against S_m.

Table 1. **The leak, and what it is.** AUROC and CUR of native confidence for invented against clean observed pixels (blur and noise, 18 scenes × 5 seeds × 2 radii), and, shaded, against an identical patch that another view sees (matched split). Intervals in the supplement.

	DTU (lab rig)				CO3Dv2 (phone)			
	vs clean		vs observed patch		vs clean		vs observed patch	
model	AUROC↑	CUR↓	AUROC↑	CUR↓	AUROC↑	CUR↓	AUROC↑	CUR↓
VGGT-1B	0.90	0.24	0.55	0.80	0.68	0.65	0.49	0.85
Pi3	0.91	0.28	0.49	0.83	0.89	0.27	0.49	0.81
Pi3X	0.81	0.53	0.47	0.87	0.73	0.57	0.50	0.81
MapAnything	0.74	0.62	0.51	0.84	0.58	0.81	0.49	0.84
Depth Anything 3	0.71	0.70	0.43	0.85	0.62	0.80	0.48	0.84

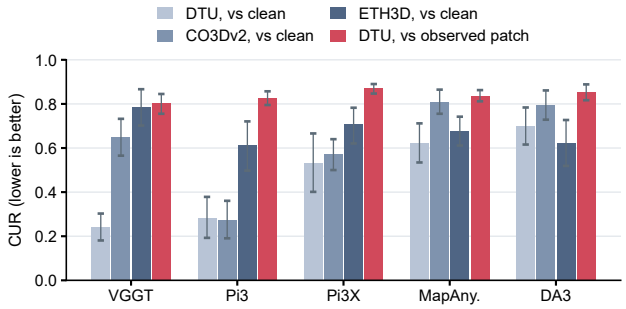


Figure 3. **Every model leaks, and more against the matched control.** CUR of native confidence (95% intervals over scenes) against clean pixels on three datasets, and against an identical observed patch (red). Lower is better.

Data. *DTU* [8]: the 18 scans of the MVSNet validation split, 10 views spread over the 49-camera rig; the 22 scans of its test split are a second real-occlusion set. *CO3Dv2* [24]: 18 phone orbits, the two depth-carrying sequences of nine categories, with COLMAP depth. *ETH3D* [25]: nine building-scale DSLR scenes with laser-scanner ground truth. *Phototourism* [10]: nine landmarks of the Image Matching Challenge 2020, Internet photographs with people in front of the monument and unordered focal lengths. Every threshold is frozen on DTU. Three backbones list CO3Dv2 in their training data; none lists DTU or ETH3D.

Models. VGGT-1B [29], Pi3 and Pi3X [32], MapAnything [13] and Depth Anything 3 [17], each with its own depth and confidence, at the same views and resolution. A Gaussian-splatting model [9], scored with its opacity, gives the same answer (supplement).

4. Native confidence reads the patch

The leak. Table 1 and Figure 3: every model lets a large share of invented geometry through, from CUR 0.24 for VGGT to 0.70 for Depth Anything 3 on DTU. Three tiers emerge: VGGT and Pi3 drop their confidence inside the hidden region yet pass one in four unsupported pixels; Pi3X passes one in two; MapAnything and DA3 are close to blind.

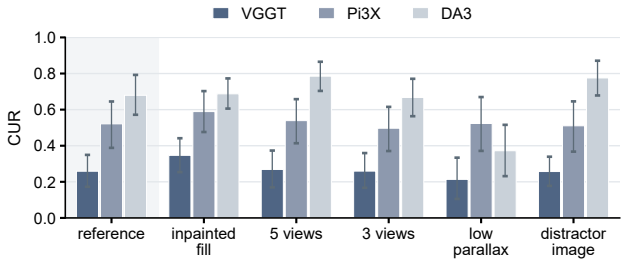


Figure 4. **Stress tests on DTU** (blur, radius 0.18, seeds 0 and 1). A seamless inpainted fill raises the leak; fewer views or a distractor image leave it unchanged.

The gray fill, an obvious patch, gives the smallest numbers (down to 0.053). The invented depth is accurate (ABSREL_U 0.011 to 0.022): the failure is that imagined cannot be told from observed, not that the imagination is poor. MapAnything is the exception, with 29% of its unsupported pixels confident and wrong by more than 10%. After Holm correction VGGT and Pi3 are indistinguishable ($p = 0.41$) and both leak less than the other three ($p \leq 0.0015$).

Phone captures change the ranking. On CO3Dv2 Pi3 is the only model whose leak transfers from the rig (CUR 0.27 on both); VGGT, joint best on DTU, leaks 2.7× more, and MapAnything and DA3 pass four in five. The leak is largest on the dataset three of the models were trained on: seeing CO3Dv2 did not teach their confidence to notice a missing surface.

What the confidence reads. Shaded blocks: against an identical patch that another view does see, native confidence separates nothing. AUROC_m is 0.43 to 0.55 on DTU and 0.48 to 0.50 on CO3Dv2, and a threshold that keeps 90% of the observed patch keeps 80% to 87% of the imagined one. Clean pixels against the observed patch score 0.75 to 0.92: the original AUROC is the confidence dropping under a patch, whichever label it carries. Every model's original AUROC exceeds its matched one by 0.24 to 0.43, so the matched numbers are the ones to read.

Not the fill, not the views, not cheap fixes. An inpainted fill without seams raises the leak for VGGT (+0.088, $p = 0.031$) and Pi3X (+0.068, $p = 0.005$), so Table 1 is a lower bound for a natural-looking occluder (Figure 4). Dropping from 10 to 3 views leaves VGGT and Pi3X unchanged ($p > 0.4$): fewer views to cross-check do not make the models less sure, so native confidence follows single-view appearance. DA3 differs, its leak rising with a distractor and halving at low parallax, consistent with a confidence that tracks appearance consistency across views. Variance over four perturbed runs is far worse than native confidence (AUROC 0.67 and 0.55 against 0.92 and 0.83) and reprojection with the model's own cameras is at chance on the

Table 2. **Single-view geometry on clean views,** mean over four backbones (VGGT-1B, Pi3X, Pi3, DA3; per-backbone numbers and intervals in the supplement). Our heads are tinted, the best value per column is bold. No head saw these scenes.

	DTU		CO3Dv2		ETH3D		Photot.	
detector	AUROC↑	CUR↓	AUROC↑	CUR↓	AUROC↑	CUR↓	AUROC↑	CUR↓
native confidence	0.82	0.49	0.67	0.80	0.76	0.50	0.82	0.46
image-border cue	0.70	0.73	0.45	0.90	0.73	0.63	0.73	0.65
per-image control	0.85	0.40	0.45	0.89	0.66	0.72	0.72	0.68
reprojection, own cameras	0.94	0.13	0.85	0.40	**0.97**	**0.07**	0.82	0.34
support head, GT labels	**0.99**	**0.02**	**0.92**	0.24	**0.97**	0.08	**0.93**	0.21
support head, label-free	**0.99**	0.03	**0.92**	**0.23**	**0.97**	0.09	**0.93**	**0.19**

patches (0.49, 0.48): the invented surface is stable and cross-view consistent.

Takeaway. The shipped confidence detects the look of a patch. A detector of unobserved geometry must pass a test where appearance cannot explain the answer.

5. Method: a support head that abstains

Architecture. The backbone (VGGT, Pi3X, Pi3 or Depth Anything 3) stays frozen. The head taps the tokens its depth or confidence head reads (for VGGT and DA3 four cross-view layers, 2048-d at 14×14 patches), applies LayerNorm and a 1×1 projection per layer, sums, applies two 3×3 convolutions and returns one logit per patch (Figure 5). The score is $1 - P(\text{imagined})$; three bands fixed in advance tell the user *observed* (> 0.7), *abstain* or *imagined* (< 0.3).

Labels without ground truth. The backbone runs once on clean views; each predicted point is projected with the predicted depth and cameras into the other views and counted where it lands on their predicted surface. No other view: imagined. Two or more: observed. This is the reprojection test used as a teacher, so the head must generalise beyond it to be worth having. Ground-truth visibility gives the upper bound. *Matched labels* use the three-patch protocol, with the observed patches as negatives so that a patch cue is penalised. All heads: class-balanced binary cross-entropy, 1500 AdamW steps (learning rate 10^{-3}, cosine schedule), 3 to 10 random clean views per step on the 79 DTU training scans, disjoint from every benchmark scene; ten minutes on one H200, hyperparameters fixed before any benchmark scan was evaluated.

Controls. The decisive control is the *per-image head*: same head, same labels, on tokens before cross-view attention. Under matched labels it must stay near chance, or the pipeline leaks. Further: a linear probe per layer, image-border and depth-edge cues, extra seeds, and a provenance variant with cross-view attention that says *which* view corroborates a pixel (supplement).

6. Experiments

Four datasets the head never saw. Figure 6 and Table 2: trained on DTU training scans only, the label-free head separates single-view from supported geometry at AUROC 0.98 to 0.99 on DTU, 0.88 to 0.94 on CO3Dv2, 0.95 to 0.98 on ETH3D and 0.92 to 0.94 on the landmark photographs, against 0.55 to 0.89 for native confidence. It is within 0.02 of a head trained on ground-truth visibility, beats its teacher, reprojection, on DTU, CO3Dv2 and the landmarks, and ties it on ETH3D, where adjacent DSLR views make reprojection as good as any head. On the landmark photographs, where people stand in front of the monument and the model's own cameras are least reliable, the head beats reprojection by 0.11 to 0.13. Figure 7 shows what this looks like.

Where the signal lives. The per-image control falls below chance on CO3Dv2 and ETH3D: without cross-view evidence the head learns the frame edge, which marks single-view pixels on the rig and nowhere else. A linear probe on one layer at a time (Figure 8) shows support becoming linearly decodable in the second half of the aggregator, and a probe on all four tapped layers is within 0.015 of any head we trained: the information is in the features, and the head is a reader. A head trained on the benchmark's fills instead learns the fill (AUROC 1.000, and so does its per-image control) and scores below native confidence on real single-view geometry: masking labels need the control.

A larger head reads no more. If the head did the work, capacity would help. On real occlusion (DTU and ETH3D, 3, 5 and 10 views, VGGT and Pi3X) we compare three readers on the same features and labels: a linear probe (0.5 to 2.1 M parameters), our head (1.7 to 3.3 M) and a cross-view transformer head with two global-attention blocks over the tokens of all views (2.1 to 3.7 M). The transformer head is better in none of the 12 settings (-0.041 to $+0.006$ AUROC), and the linear probe is within 0.03 of our head in 11. Our head beats the per-image control in 11 of 12 ($+0.05$ to $+0.13$). The support signal is already linearly readable in the frozen features; the head is light because nothing more is needed.

The matched head. Trained on the three-patch protocol on all four backbones (Figure 9), the per-image control stays at 0.56 to 0.61 AUROC_m (bound 0.60, set before training), so the pipeline does not leak. The head separates the imagined patch from the observed one at AUROC_m 0.69 to 0.77 against 0.43 to 0.55 for native confidence ($p \leq 0.0003$ on every backbone), the label-free head within 0.04 of the ground-truth one, and holds on CO3Dv2 without retraining (0.62 to 0.69 against 0.54 to 0.58 for its control). Its accuracy rises with the evidence, from 0.68 to 0.748 with one clean view at low parallax to 0.80 to 0.92 with three or more: the signature of a detector that reads cross-view support, not appearance. A single low-parallax view is the hardest case, and a fine-tune that oversamples it does not move it (0.71). The check earned its place: a first placement that ignored

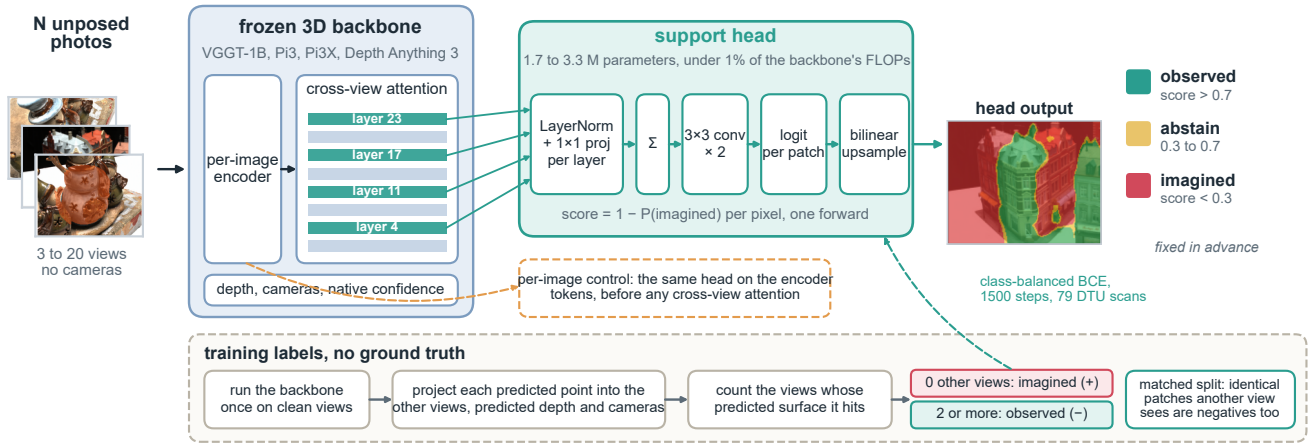


Figure 5. **The support head.** A frozen backbone's cross-view layers feed a small head that returns, per pixel, whether the output is corroborated by other views, and abstains in between. Its training labels come from the backbone's own prediction: each predicted point is reprojected into the other views and counted. No ground truth, no change to the backbone, one extra forward under 1% of its cost. The per-image control is the same head on tokens taken before any cross-view attention.

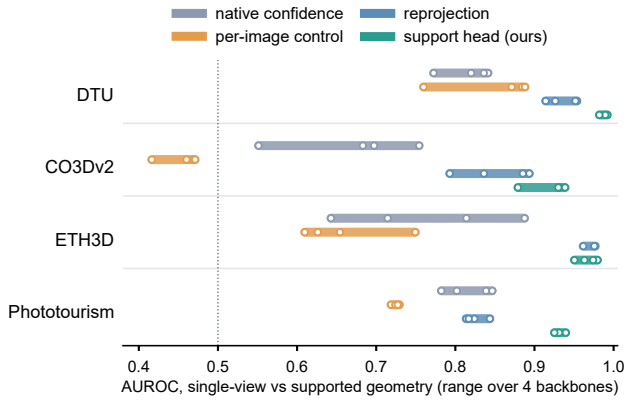


Figure 6. **Four datasets the head never saw.** AUROC for single-view against supported geometry on clean views (3, 5 and 10), range over four backbones (dots). The per-image control falls below chance off the lab rig. Full table in the supplement.

texture let the observed patch fall on the rig's backdrop and the control rose to 0.62, so the run was repeated.

Real occlusion against dense references. No patch and no fill: the backbone runs on 3, 5 or 10 real views, and each predicted point is matched to a dense reference that records which surfaces the input views saw (Figure 10). On DTU the reference is all 49 rig views; about 1 to 1.7% of the predicted points lie on surfaces no input view saw with three views, 0.5 to 0.8% with five and 0.2 to 0.4% with ten. Where that share reaches 0.5%, on the 18 benchmark scans and on 22 further test scans, the head beats native confidence in 14 of 16 settings: for every backbone at five views ($+0.08$ to $+0.13$ AUROC, $p \leq 0.0004$) and for Pi3X, Pi3 and DA3 at three ($+0.08$ to $+0.14$, $p \leq 0.005$); VGGT at three views gains $+0.02$ and $+0.04$ without significance. At ten views the share is too small for a claim, and the gain is largest ($+0.09$ to $+0.18$). On CO3Dv2 phone orbits, with the MVS depth of every frame as reference, the head does not beat native confidence (-0.11 to $+0.07$): its DTU-trained reading of support does not carry over to object-centric phone video. ETH3D's laser depth leaves about a third of the camera-visible surface without depth, too much for the observed test, so ETH3D is not used here (supplement).

Which view saw it. The provenance variant (Figure 12) ranks the views that see a point above those that do not at prov-AUROC 0.93 to 0.96 on DTU and 0.83 to 0.88 on CO3Dv2 with no camera, the level of reprojection there, and trails reprojection on building scans and landmarks, so the convolutional head stays the method of record (supplement).

Adapting without labels. Trained label-free on half of a target dataset and scored on the other half, the head gains where the backbone's own geometry is reliable (Pi3X on CO3Dv2 and ETH3D, $+0.025$ and $+0.027$) and loses where its teacher is weak (VGGT on CO3Dv2, -0.09); the DTU-trained head is the safe default.

Downstream: a certified reconstruction. The head keeps almost no single-view geometry until high coverage (Figure 11a). With the bands fixed in advance, the head marks 57% to 87% of the pixels *observed*, of which 0.5% to 5% are single-view; a threshold set on other scenes lets 1% to 2% of single-view pixels through on DTU against 57% to 83% for native confidence. Keeping the 80% of points the head trusts most raises the accuracy of the fused

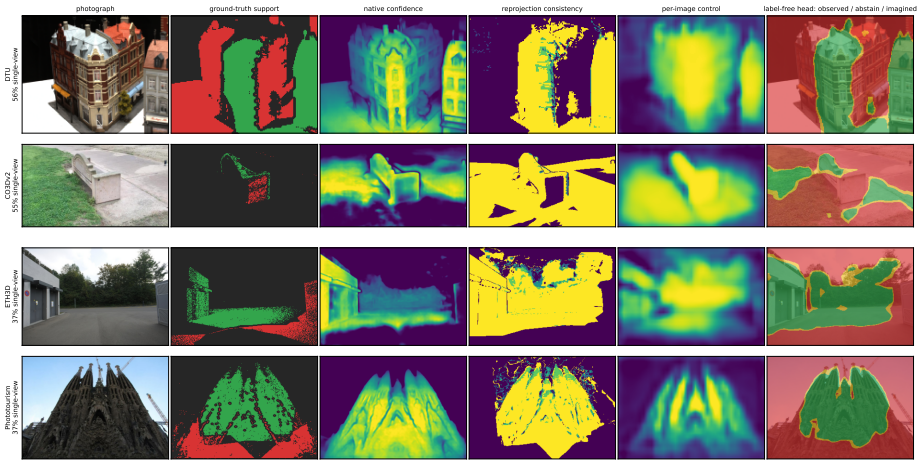


Figure 7. **What the label-free head says on unseen captures.** Per dataset: photograph; ground-truth support (red: no other view sees it, green: two or more do); native confidence; reprojection; per-image control; the head's bands (green observed, yellow abstain, red imagined). On DTU native confidence lets half of the single-view pixels through a 90% filter, the head 1.7%.

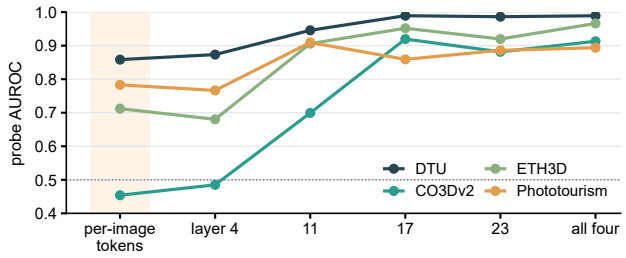


Figure 8. **Where the signal lives.** A linear probe on one VGGT layer at a time. Per-image tokens (shaded) carry only a frame-edge cue that fails off the rig; support becomes linearly decodable from layer 17 of 24.

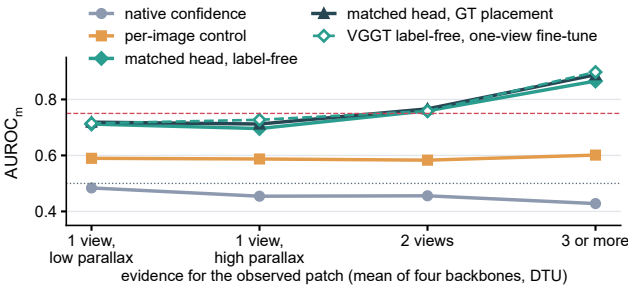


Figure 9. **The matched head reads evidence.** AUROC$_m$ of the imagined against the observed patch, by how many clean views see the observed one (mean of four backbones). Native confidence and the per-image control stay near chance; the head improves with evidence. Dashed: the 0.75 target and the one-view fine-tune.

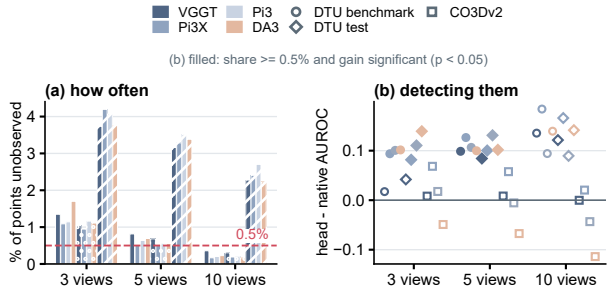


Figure 10. **Real occlusion against dense references** (DTU: all 49 rig views; CO3Dv2: MVS depth of every frame). (a) Share of predicted points on surfaces no input view saw; a claim needs at least 0.5% (dashed). (b) The head's AUROC gain over native confidence on those points; filled where the gain is significant. ETH3D is not used for this test (supplement).

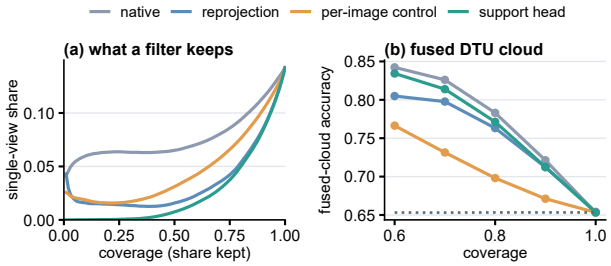


Figure 11. **What abstaining buys** (VGGT, DTU). (a) Single-view share of the pixels a filter keeps, by coverage. (b) Accuracy of the fused point cloud when the least trusted points are dropped. Native confidence tracks error, the head tracks support.

DTU cloud from 0.65 to 0.77 (Figure 11b); at that coverage the head keeps half as much single-view geometry as native confidence (2.5% against 4.6% of the points) at nearly the same accuracy (0.77 against 0.78). Abstaining thus yields a cloud that is as accurate and twice as well corroborated. The two scores measure different things: native confidence ranks depth error, the head ranks support. Averaging them

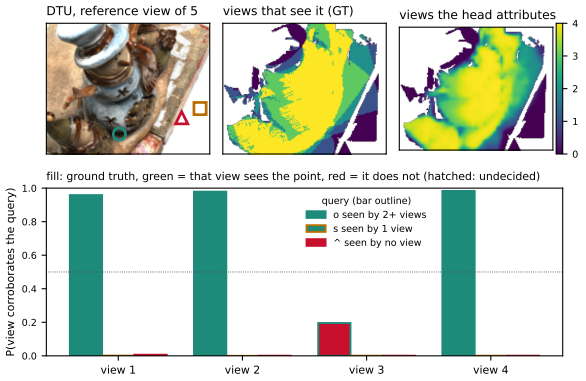


Figure 12. **Which view saw it** (provenance variant, VGGT, an unseen DTU scan). Top: views that see each pixel and views the head attributes. Bottom: the head's probability per view for three query pixels; bar fill is the ground truth.

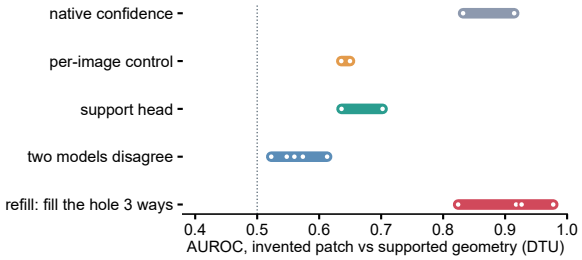


Figure 13. **The limit.** On the benchmark's own patch, which every view shows, no cross-view test separates the invented surface; refilling the hole does. Native confidence scores high only because it reacts to the patch (Table 1).

changes fused F1 by less than 0.01, and with three views, where real unseen geometry is most common, the head's cloud is about as accurate as native confidence's (-0.02 to $+0.04$ at 80% coverage, DTU and ETH3D) while keeping 18% to 39% less single-view geometry in every setting. A user can apply one filter for accuracy and the other to certify what was seen.

The limit. No cross-view test fires on a surface every view shows (Figure 13). On the benchmark's own patches the visibility heads score 0.70 to 0.75 AUROC, and disagreement between two different backbones 0.52 to 0.61: five models of three families invent the same surface. That surface is not fixed by the prior, though. Filling the same hole three ways moves it, and the spread separates the patch at AUROC 0.82 to 0.98 on DTU: the model reads the hole and draws what its pixels suggest.

Takeaway. The backbone knows which of its geometry has multi-view support. A small head reads it without labels; what no cross-view test can read is an invention every view agrees on.

7. Limitations and conclusion

Limitations. The head certifies multi-view agreement, not truth: a surface every view agrees on passes, whether a camera saw it or the prior drew it. Real unseen geometry is rare when views are many (0.2 to 0.4% of the points at ten DTU views), so the real-occlusion claim is made at three and five views. On CO3Dv2 phone orbits the DTU-trained head does not beat native confidence on real occlusion, and it adds no fused-cloud accuracy over native confidence. ETH3D's laser depth leaves about a third of the camera-visible surface without depth; the same gaps affect its single-view labels, which we read as the noisiest of the four datasets. CO3Dv2 depth is COLMAP with 1% to 3% noise and three backbones saw CO3Dv2 in training; the landmarks' ground truth has holes; scenes are static. A single low-parallax view remains the hardest matched case.

Conclusion. Feed-forward 3D models mostly do not say when they have not seen a surface, and their confidence reacts to appearance. Their cross-view tokens do carry the answer: a light head trained on the backbone's own agreement reads it on captures it never saw and on the real unseen surfaces of 40 scans, and more capacity reads no more. The next problems are the invention every view agrees on and support detection that transfers to casual video.

LLM Usage Statement

A general-purpose LLM assistant was used, under the authors' direction, for parts of the code (the benchmark runner, the model wrappers, the run scripts and figure scripts), for literature lookup, and for editing the text of this paper, including grammar. The study design, the definitions and metrics, the choice and freezing of every hyperparameter, the execution and checking of every experiment, the verification of every reported number against its logged run, and the conclusions are the authors' own, and the authors are responsible for all content. The assistant was not used to generate results, figures from unlogged data, or references; every citation was checked against the published record. The tool is named in the camera-ready version.

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